

Master's Thesis

A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV

Graduate School, Korea University
Department of Mechanical Engineering
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Abstract (English rendering of the Korean abstract)

Disaster-response technologies utilizing robots are rapidly advancing. Drones, in particular, are being used for spatial exploration and victim search in indoor and outdoor disaster environments because of their high mobility and broad exploration range. In situations where direct human access is difficult, teleoperation technology allows drones to be controlled remotely. In such cases, the usability of the controller and the performance of the user interface for conveying data collected by the drone directly affect mission execution.

This study proposes a new controller and mapping method that enable more intuitive drone operation. A monocular-camera-based user interface was also developed for lightweight drones. This interface includes 3D spatial reconstruction, obstacle segmentation, and predicted flight-path visualization. These functions support operators in understanding spatial information in real time and making effective flight decisions.

Simulation-based experiments confirmed that the proposed interface system improves the stability of remote drone operation and reduces operator fatigue during mission execution. In particular, usability improved by 28% compared with the conventional RGB-video-based user interface, and workload measurements also showed meaningful improvements in every category. These findings demonstrate that this study can contribute to improving the efficiency of disaster-response mission execution using drone technology.

Keywords: Drone, Teleoperation, Controller, User Interface

A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV

by Seungyeop Lee

Department of Mechanical Engineering

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Abstract

Disaster response technologies utilizing robots are rapidly advancing, and drones, in particular, are being employed for spatial exploration and victim search missions in both indoor and outdoor disaster environments due to their high mobility and broad exploration range. In situations where direct human access is difficult, teleoperation technology enables remote control of drones, and in such cases, the performance of controllers and user interfaces for conveying data collected by the drone directly affects mission success.

This study proposes a new controller and mapping methodology to enable more intuitive drone operation. Additionally, a monocular camera-based user interface was developed for lightweight drones, incorporating features such as 3D spatial reconstruction, obstacle segmentation, and predicted flight path visualization. These functionalities support operators in understanding environmental information in real-time and making effective flight decisions.

Simulation-based experimental results confirmed that the proposed interface system improves the stability of remote drone operation and reduces operator fatigue during mission execution. Notably, the system demonstrated a 28% improvement in usability compared to conventional RGB video-based user interfaces, and task load measurements showed significant improvements across all categories. This study demonstrates that the proposed system can contribute to enhancing the efficiency of disaster response missions using drone technology.

Keywords: Drone, Teleoperation, Controller, User Interface

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Chapter 1. Introduction

1.1 Research Background

Disasters caused by natural hazards and human factors result in economic damage, destroy national infrastructure and buildings such as apartment complexes, and create social and economic problems. Various efforts are being made to minimize this damage and respond effectively to such situations, with disaster response using robots attracting particular attention [1, 2]. Robots used for disaster response and facility exploration are classified as Unmanned Ground Vehicles (UGVs), Unmanned Marine Vehicles (UMVs), and Unmanned Aerial Vehicles (UAVs), depending on their operating environment and required functions. Among these, UAVs, or drones, are suitable for monitoring and exploring extensive areas of indoor and outdoor disaster sites and are used in a variety of situations [3]. Because drones offer relatively low cost and a high degree of freedom of movement, they are useful for inspecting and exploring the interiors of irregular structures or searching for victims in disaster-response operations where human access is difficult [4]. Indeed, following the Fukushima nuclear power plant accident in Japan, drones were used to capture gamma-ray images and complete missions that could not be performed from the ground [5, 6].

Robot teleoperation technology is needed to perform missions using robots in environments where people cannot be deployed. This technology allows an operator to manipulate a robot at a remote location according to the operator's intentions. Despite advances in autonomous navigation technology, it continues to play a central role in mission execution [7]. In situations requiring an immediate response, such as avoiding unexpected obstacles or searching for victims, direct operator control is essential, and robot sensors are strongly affected by external factors such as light and radio signals [8]. In these circumstances, situational awareness is very important when an operator performs a mission through teleoperation [9]. Situational awareness refers to the operator's understanding of the environment based on information collected through the robot. It is a key element of teleoperation because it directly affects spatial perception, situation assessment, and fatigue during mission execution [10]. In drone operation, operators perceive the drone's surroundings solely through images transmitted within a limited camera field of view or sensor data displayed on the user interface. This increases the likelihood of incorrect decisions and leads to high operator fatigue. Drone size must also be considered in teleoperation using drones. Indoor disaster-response drones must fly through confined spaces, which limits their size [11]. The resulting reduction in allowable payload makes it unsuitable to mount large-capacity batteries, limiting flight time. Therefore, performing extended missions in disaster situations requires reduced energy consumption through drone weight reduction and the development of a mission-optimized user interface [12].

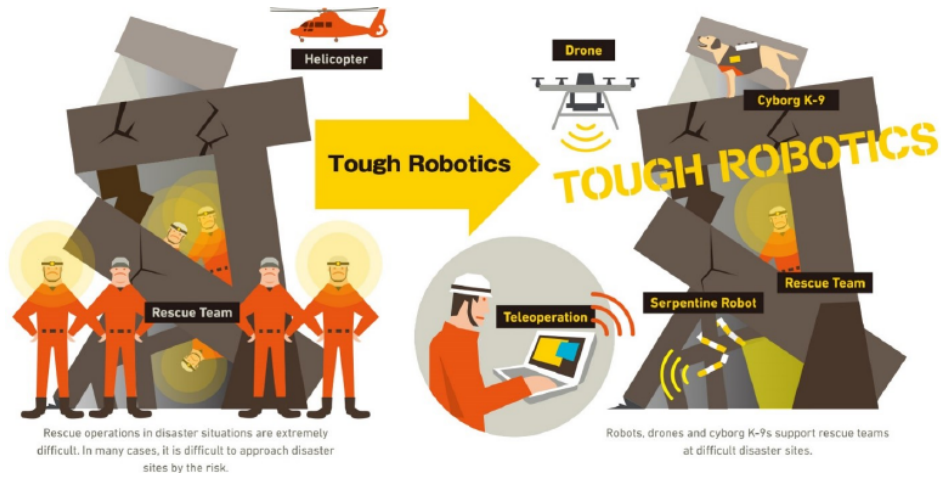


Figure 1.1. Teleoperation for Disaster Response

Controllers in current drone systems are mainly gamepad-style devices, and operators perform missions using two small joysticks [13]. However, these controllers are not intuitive to operate, making operating errors more likely when used by inexperienced operators. This limits the range of drone applications and raises concerns about reduced reliability in emergency situations. According to the Fire and Disaster Headquarters, drone accidents caused by inadequate piloting account for 33% of all failures, a problem identified as having a significant impact on the efficiency and safety of drone operation.

Various studies have been conducted to address these problems. To improve controller intuitiveness, researchers have studied wand-style controllers that match drone motion to controller motion and human-drone interaction that controls drones by sensing hand movements [14, 15]. However, precise control is difficult with these new controller designs, limiting their use in tasks such as flight through confined spaces and obstacle avoidance.



Figure 1.2. Drone Controller Using Motion Matching

Studies using LiDAR sensors for indoor environment perception and studies of 3D spatial reconstruction by small drones using monocular cameras have also been conducted [16, 17]. Because LiDAR sensors provide high accuracy, they can be used to analyze the geometry and dimensions of a space precisely. However, they have the disadvantages of high power consumption and demanding computational requirements. In contrast, a monocular camera enables drone weight reduction, but it cannot provide metric distance information, limiting its use as spatial-perception information in 3D reconstruction [18].

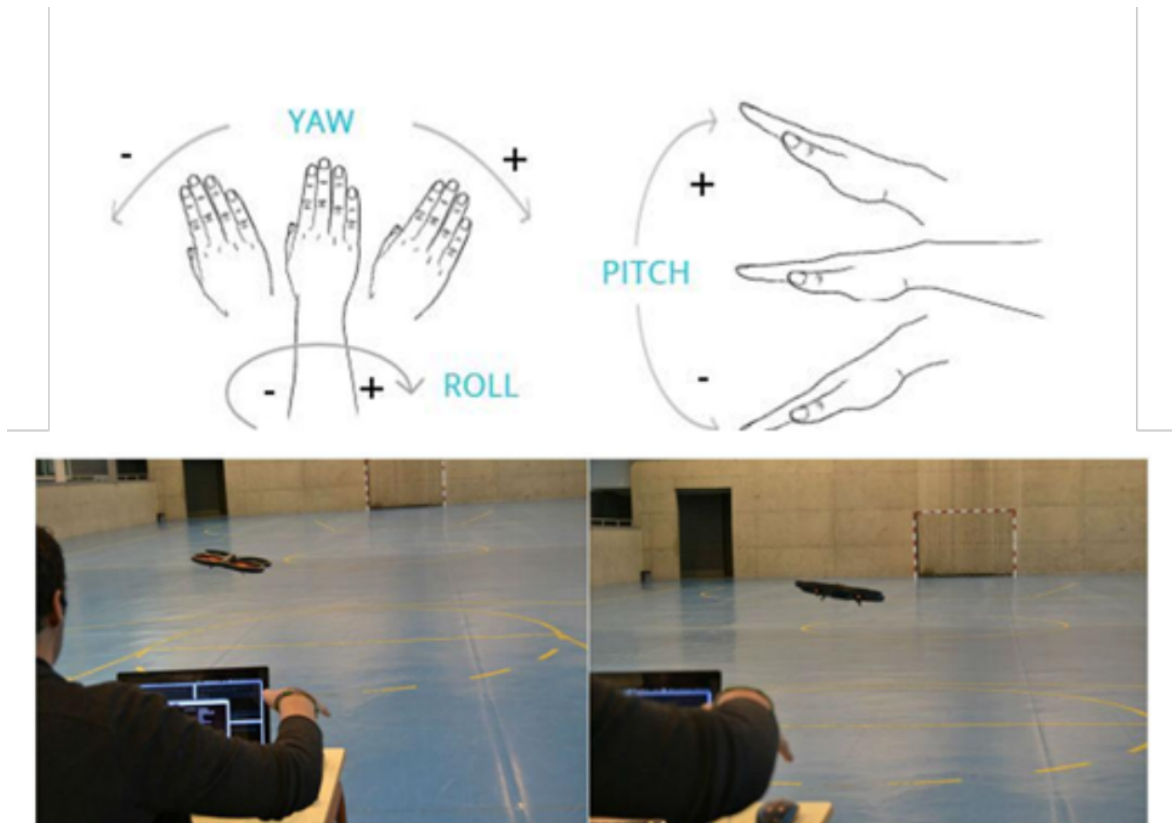


Figure 1.3. Drone Control System Using Hand Motion

The preceding analysis indicates that disaster-response drone control systems must consider the following factors. First, a controller that enables intuitive operation must be designed. A non-intuitive controller causes operating errors among inexperienced operators, resulting in unstable control of drones in complex environments. Second, drone batteries and sensors must be made lightweight. Drones deployed in confined spaces are small and have limited payload capacity, so high-performance sensors with substantial power consumption are unsuitable for extended flight. Finally, a user interface that can improve the operator's situational awareness is needed because the operator's awareness of the remote environment directly affects the outcome of remote manipulation tasks.

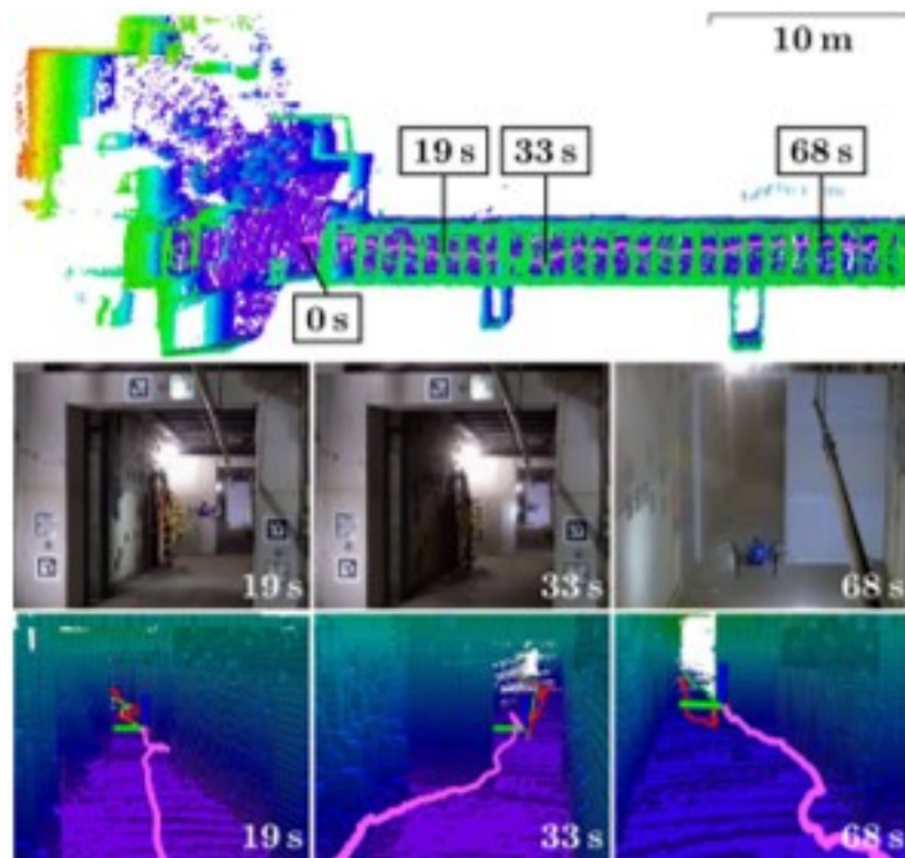


Figure 1.4. Indoor Spatial SLAM Using a LiDAR Sensor

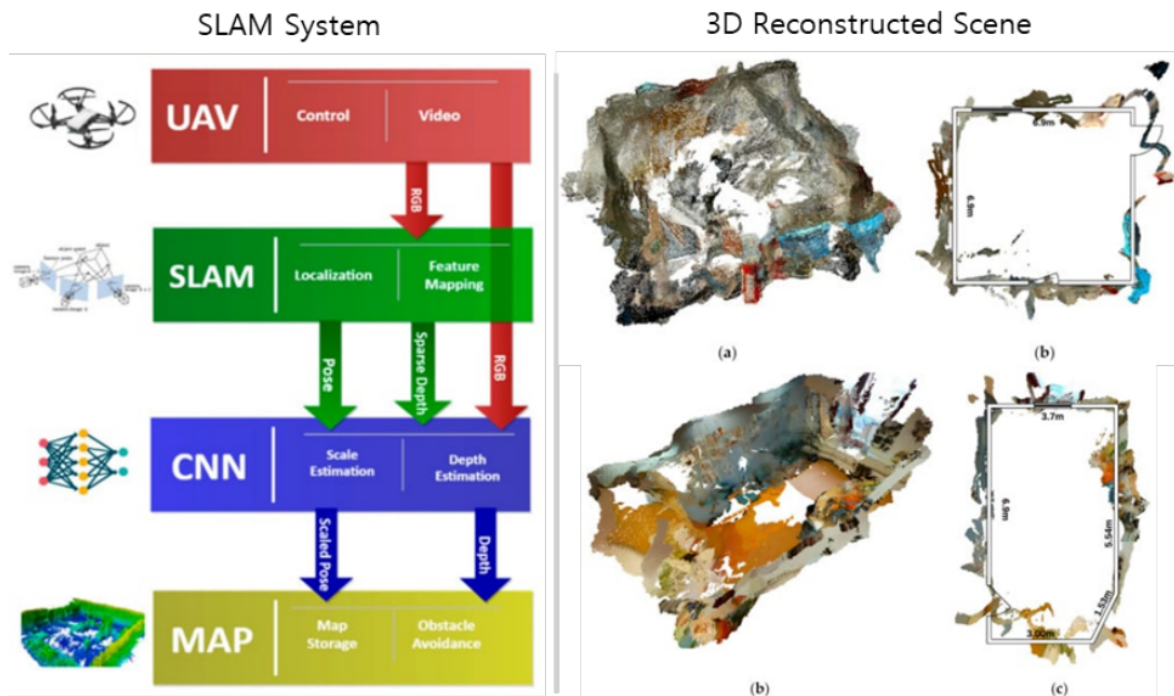


Figure 1.5. 3D Reconstruction of Indoor Space Using a Monocular Camera

1.2 Research Objectives and Scope

This study aims to develop a controller that enables intuitive operation and an operator-assistance user interface that supports real-time situational awareness. Controller mapping that reflects the characteristics of drone motion can reduce operating errors and improve flight stability. A user interface that assists situational awareness is also developed to improve system usability and reduce operating fatigue during mission execution. To this end, a monocular camera is selected as the sensor with the goal of designing a user interface based on lightweight sensors, and a user interface using this camera is proposed. The proposed system is validated through user experiments. Ultimately, using the proposed controller and user interface in disaster environments can improve operators' mission performance. Figure 1.6 provides an overview of the system developed in this study.

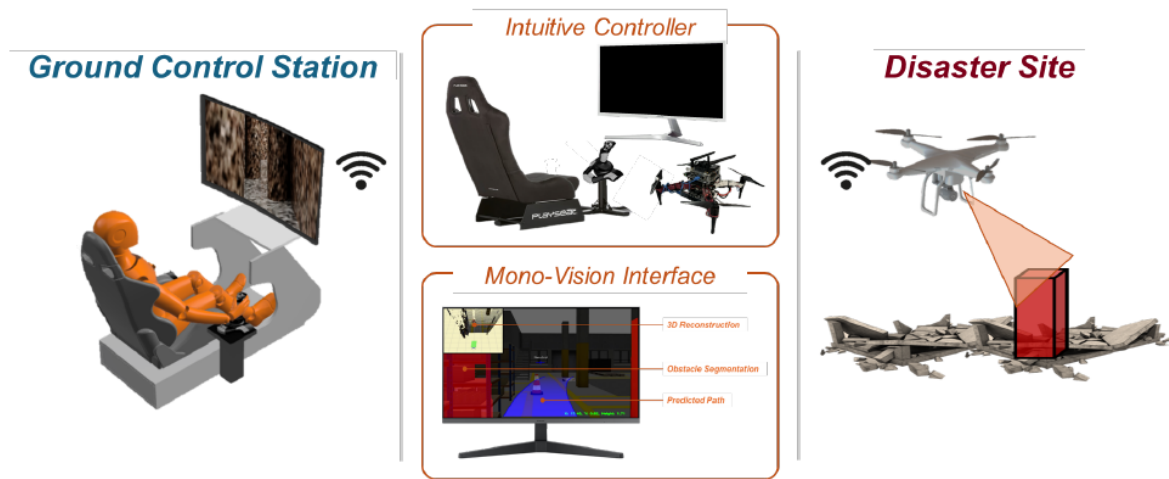


Figure 1.6. Overview of the Drone Teleoperation Controller and User Interface

The chapters are organized according to the following stages. Chapter 2 explains the control system developed in this study, including an analysis of drone motion, the limitations of existing systems, the required controller elements, and the controller design and mapping method. Chapter 3 addresses the user interface design developed in this study, explaining the key elements of a user interface for disaster-response drones and each component of the developed interface. Chapter 4 describes experiments using the developed control system and user interface, presenting the experimental scenarios and methods and the results obtained. Finally, Chapter 5 presents the overall conclusions and key findings of this study.

Chapter 2. Controller Design

Chapter 2 describes the characteristics of drone motion and selects a controller suitable for intuitive drone operation. It then explains a controller-mapping method, inspired by helicopter controls, that allows drone position-control functions to be performed intuitively.

2.1 Characteristics of Drone Motion

Drone motion is controlled by differences in the rotational speeds of its multiple motors. GPS, an Inertial Measurement Unit (IMU), ultrasonic sensors, and vision sensors are used to detect changes in position and attitude in real time, and motor-control signals are generated based on this information [19, 20, 21]. As illustrated in Figure 2.1, a drone can perform six-degree-of-freedom (6-DOF) motion, including horizontal flight forward, backward, left, and right; altitude adjustment through ascent and descent; and heading control through rotation [22]. These characteristics allow a drone to hover, flying while maintaining a fixed position, which is an important consideration in drone controller design.

2.2 Controller Design Considerations

Because drone operation depends heavily on the operator's skill, the risk of a crash may increase when a non-expert operates the drone [23]. To prevent this, intuitive and user-friendly controller design and control mapping are needed. Most current drone controllers are gamepad-style devices using two joysticks, as shown in Figure 2.2 [24]. Typically, of the two joysticks arranged side by side, the left joystick controls altitude and heading, while the right joystick controls horizontal movement. However, whereas a drone moves in three dimensions, the controller provides only two-dimensional planar inputs. The resulting lack of intuitiveness can increase operator fatigue [25].

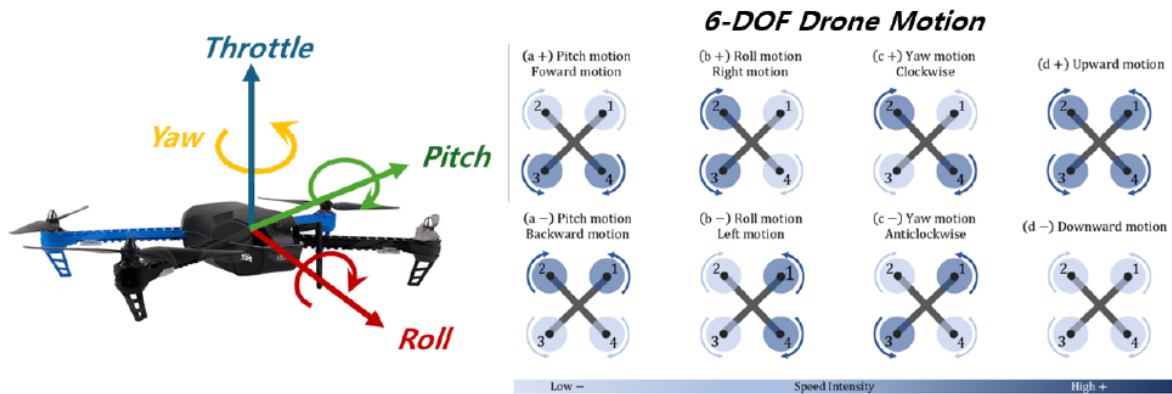


Figure 2.1. 6-DOF Drone Motion



Figure 2.2. Gamepad-Style Drone Controller

To control the drone's six-degree-of-freedom motion intuitively, this study drew on helicopter control-design principles [26]. Drones and helicopters are classified as rotary-wing aircraft and share motion characteristics such as vertical takeoff and landing, hovering, forward and backward flight, and lateral movement through similar flight dynamics. As illustrated in Figure 2.3, helicopter controls consist of the collective, cyclic, and

pedals. The collective changes all rotor-blade angles simultaneously to vary lift during flight and is responsible for altitude control. The cyclic changes rotor-blade angles individually to tilt the aircraft in a particular direction, allowing the helicopter to move forward, backward, left, or right. Finally, the pedals adjust the angle of the rotor at the tail to control the helicopter's heading. Thus, helicopter controls are designed to provide altitude control, horizontal movement, and rotational flight, with each function mapped to a different control. Given the similarities between drone and helicopter flight, helicopter control design can be applied to drone controller design. In particular, vertical takeoff and landing and hovering are important characteristics of both aircraft and should be reflected in the controller.

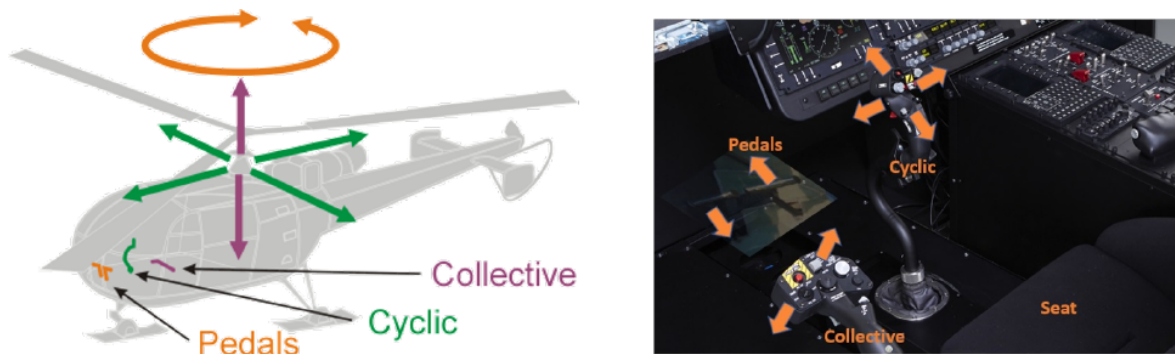


Figure 2.3. Helicopter Motion and Control Configuration

2.3 Intuitive Controller Mapping

This study selected Logitech's Extreme 3D joystick to design an intuitive controller suitable for disaster-response drones. As shown in Figure 2.4, this joystick permits continuous movement along three axes, allowing the drone's six-degree-of-freedom motion to be mapped efficiently. Wrist-operated rotation can also improve the intuitiveness of operation. In addition, its various buttons can be configured to perform necessary functions immediately. Based on the previously analyzed characteristics of drone motion, the controller and drone were mapped as follows. Forward, backward, left, and right joystick inputs were mapped to horizontal drone flight, while upward and downward operation of the centrally located lever was used for altitude control. Heading control was mapped to wrist rotation, increasing the intuitiveness of the relationship between the drone's actual motion and controller operation. The image-capture function required during missions was also mapped to a button on the left side of the controller, allowing the operator to capture images in real time at the desired moment. Finally, to prevent unintended operation while the controller is stationary and not being manipulated, a region that does not respond to small controller outputs was established to enable stable operation.



Figure 2.4. Features of the Logitech Extreme 3D Controller

To apply the mapped controller in an actual flight environment, the drone's position and heading data were collected in real time and used in the control system. In a teleoperation environment where the operator cannot directly see the drone, control must be based on real-time images collected by a camera mounted on the front of the drone. To address this limitation, the control system was designed to identify the drone's current heading from attitude data collected in real time and combine it with controller input commands, enabling accurate flight according to those commands relative to the drone's forward direction. Through the images transmitted from the drone, the operator can intuitively observe the area ahead and perform stable flight operations using the forward view shared with the drone as the reference.

Chapter 3. User Interface Design

Chapter 3 analyzes the problems of current drone control systems from the perspective of the user interface and explains a new monocular-camera-based user interface proposed to address them. The proposed interface aims to support spatial awareness of the flight area, recognition of nearby obstacles, and improved trust in the remote-control system, and includes functions for each of these purposes.

3.1 User Interface Design Considerations

Situational-awareness assistance is a key element in user interface design and strongly influences the operator's ability to understand the environment and respond appropriately [27, 28]. When operators cannot enter a space directly, they perform extended missions using only the limited information transmitted by the drone. The interface's ability to assist situational awareness therefore directly affects mission performance and safety. A poorly designed user interface during remote operation causes high operator fatigue, increasing the likelihood of errors and potentially leading to mission failure or safety problems. Accordingly, the interface must be configured to convey an appropriate amount of information while taking the operator's limited mental processing capacity into account [29].

During remote drone operation, operators depend primarily on camera images, which tends to impair their sense of distance and spatial perception. Humans perceive depth using binocular disparity, but visual information on a 2D display does not provide this information, and the camera's field of view limits the view, making it difficult to understand the drone's overall surroundings. Interpreting a space and operating the drone using only limited information increases the operator's cognitive load [30]. Indoor disaster environments have a high density of obstacles, making it difficult to perceive the space in advance. Indoor exploration and rescue operations using drones therefore require strong obstacle-avoidance capabilities. However, avoiding obstacles

with poor spatial awareness not only increases operator fatigue but also causes operating errors that can lead to collisions and crashes. To prevent this, providing a 3D image of the space can improve the operator's spatial understanding and situational awareness and reduce fatigue [31]. In addition, obstacle positions should be indicated to the operator in advance to improve flight stability. Trust in the system, based on a strong understanding of it, is also very important for improving mission performance with a drone at a remote location [32]. Visualizing the predicted flight path can improve trust in the system and give the operator a sense of security.

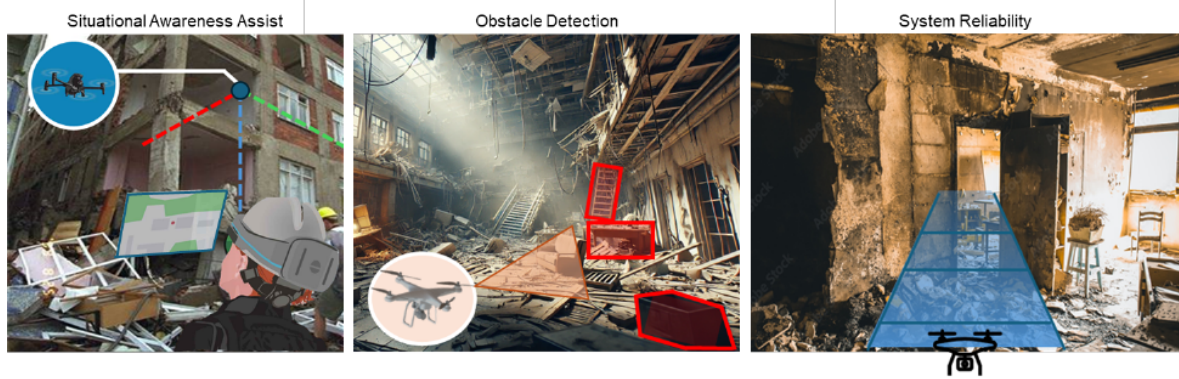


Figure 3.1. User Interface Considerations for Drone Teleoperation

As illustrated in Figure 3.1, the user interface developed in this study includes the following functions, taking into account situational-awareness assistance, obstacle-recognition assistance, and improved trust in the system. First, the space surrounding the drone is visualized in three dimensions to help the operator intuitively understand obstacles and the surrounding environment. Second, obstacle segmentation is provided to assist collision prevention and path exploration through real-time obstacle detection and visualization. Third, a predicted flight path calculated from controller inputs is displayed visually to help the operator plan paths more efficiently. These functions contribute to improved situational awareness in complex indoor disaster environments, reduced operator cognitive load, and a higher mission success rate.

3.2 3D Spatial Reconstruction

To present the space in which the drone flies to the operator as a 3D image, an image-data-processing method combining Monocular Depth Estimation (MDE) and monocular-camera-based Simultaneous Localization and Mapping (SLAM) was proposed. This method integrates the strengths of MDE and monocular SLAM to compensate for the limitations of each technique and adds odometry information to provide the operator with 3D spatial information based on metric distances. This function was configured as illustrated in Figure 3.2.

3.2.1 Monocular Depth Estimation

MDE estimates relative distances between pixels in an RGB image collected by a monocular camera using a deep learning model [33, 34]. The MDE deep learning model used in this study was the open-source MiDaS model, specifically the pretrained v3.1 DPT Swin2-L 384 model [35]. MiDaS converts the depth information of each pixel in an RGB image into an 8-bit grayscale value, regenerating the image by converting each pixel value to a relative depth value between 0 and 255, as shown in Figure 3.3. This model was trained on 12 different datasets and can estimate depth information in various environments and complex scenes.

However, without additional distance data, MDE provides only relative distance data between pixels rather than metric distances, limiting its application to measuring spatial dimensions or avoiding obstacles in real

environments [36]. Furthermore, when processing video transmitted in real time, depth values are continuously adjusted as the scene changes, resulting in a lack of consistency in the display presented to the operator. Thus, providing spatial information to the operator using a monocular depth estimation model requires additional data processing to incorporate metric distance information and stabilize the visualized information.

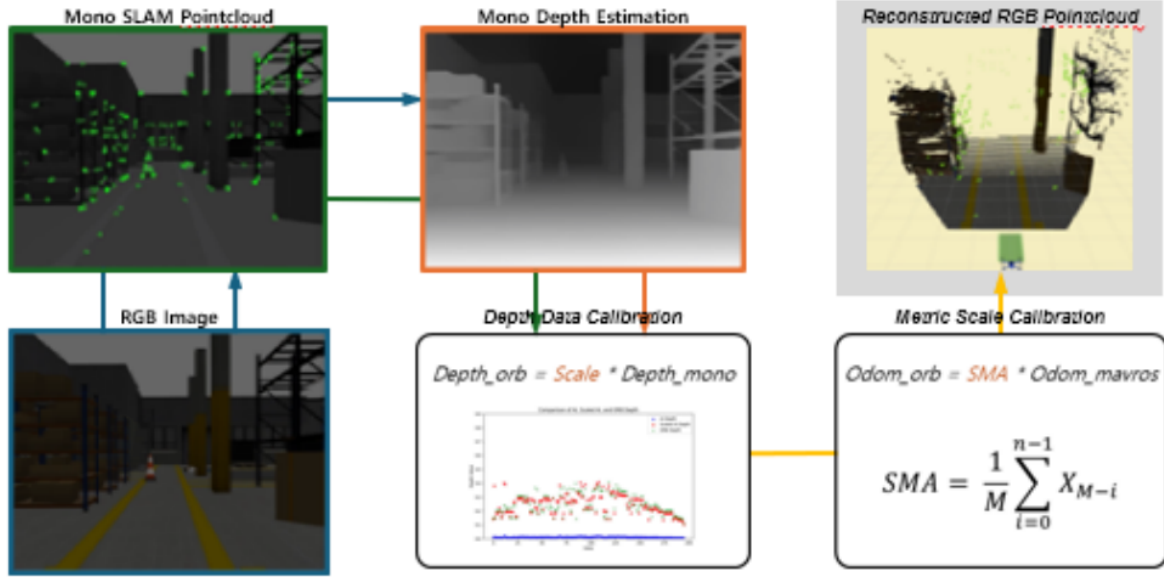


Figure 3.2. Overview of the 3D Spatial Reconstruction System

3.2.2 Monocular Visual SLAM

ORB-SLAM3, a monocular-camera-based SLAM technique, was used to collect 3D spatial information [37]. ORB-SLAM3 uses camera and IMU sensors to convert a space into a 3D point cloud while simultaneously estimating the sensor's position within the reconstructed space. Among its versions, the monocular-camera version extracts feature points from image frames based on the camera's intrinsic matrix and generates a 3D point cloud of the space by tracking changes in feature-point positions between consecutive frames, as shown in Figure 3.4. Although it has the advantage of accurately estimating the relative scale between feature points in camera images, it has the limitations of generating a limited point cloud and being unable to estimate metric distances in the reconstructed environment [38]. To overcome these limitations, the system was configured to integrate the MDE model with drone sensor data so that they complement one another.

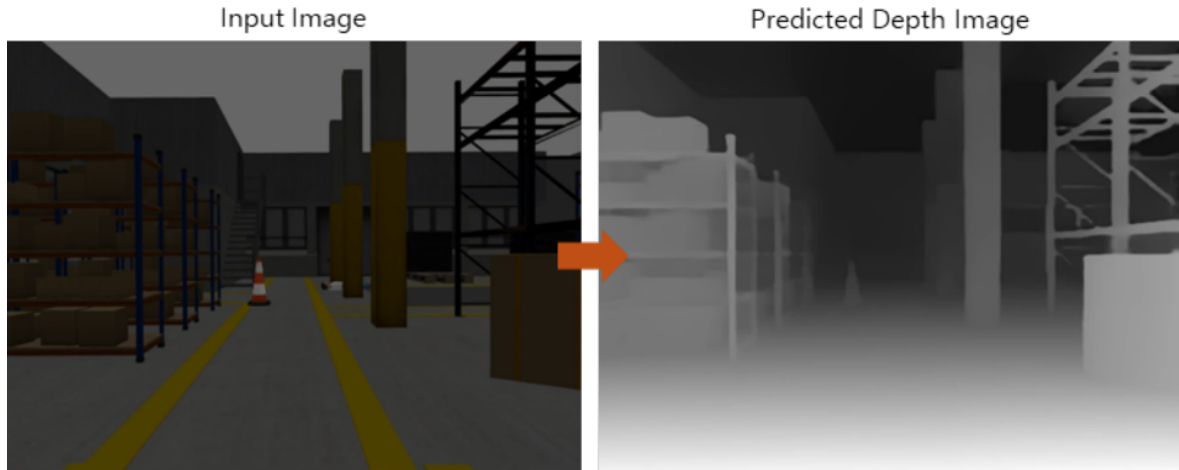


Figure 3.3. Grayscale Image Estimated by the MiDaS Monocular Depth Estimation Model

3.2.3 3D Reconstruction

To overcome the limitations of MDE and monocular-camera-based SLAM, this study proposed a two-stage scale calculation method, as shown in Figure 3.5. In the first stage, the scale between MDE relative distances and the SLAM point cloud was calculated to generate a point cloud with consistent depth information. To do so, a scale factor between the feature-point-based point cloud generated by ORB-SLAM3 and the MDE grayscale pixel values at the feature-point positions was calculated using the least-squares method. By applying this factor to the MDE depth information and converting it to a point cloud, the result was integrated with the SLAM point cloud to generate a 3D point cloud of the flight space. In the second stage, SLAM and drone Visual-Inertial Odometry (VIO) data were used to convert the relative scale into metric-distance-based data [39]. Since SLAM-based odometry does not provide movement information in metric distances, its scale relative to the drone's measured odometry was calculated and applied to the point cloud. As a result, the spatial point cloud was converted to the dimensions of the actual space, intuitively reconstructing the drone's flight space in three dimensions.

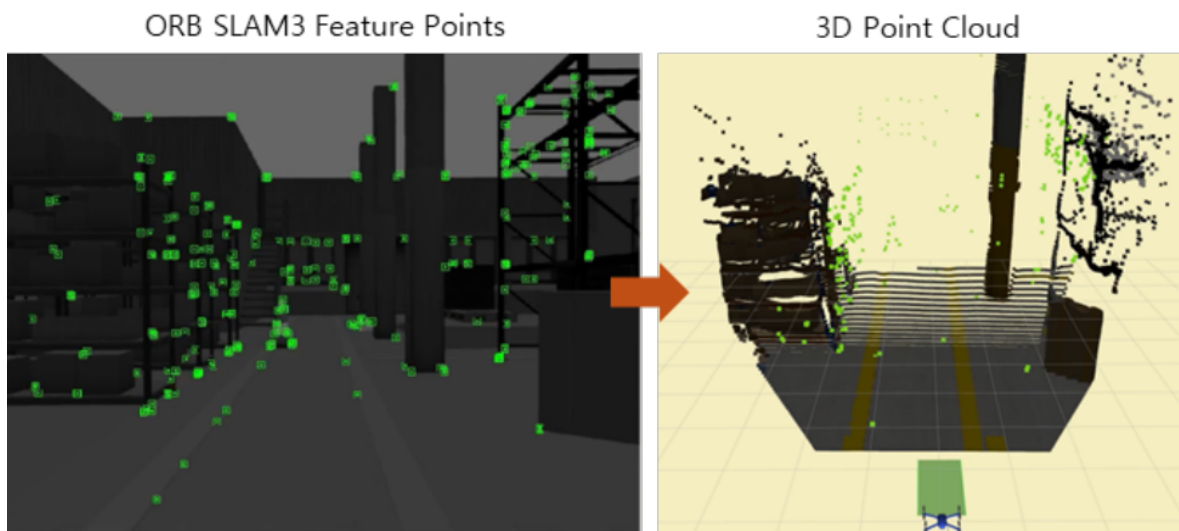


Figure 3.4. Generation of a 3D Point Cloud by ORB-SLAM3

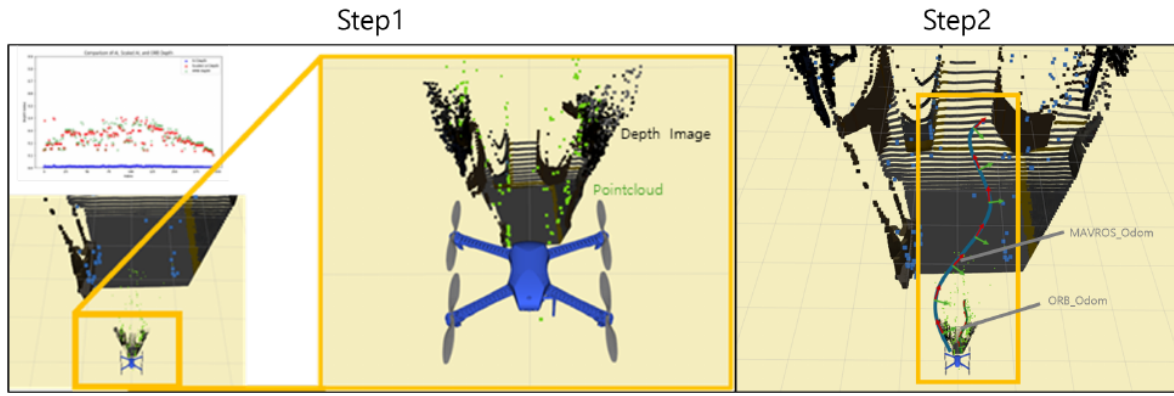


Figure 3.5. Scale Calculation for 3D Spatial Reconstruction

Through these two processes, monocular depth estimation data and the SLAM-based point cloud were combined, as shown in Figure 3.5, and scale correction using sensor-based position data was applied to generate a 3D point cloud based on metric distances. This made it possible to reconstruct the drone's surroundings in three dimensions using only a monocular camera and to present the flight space intuitively to the operator.

3.3 Obstacle Segmentation

Obstacle segmentation is necessary to prevent collisions when a drone flies in an environment containing many obstacles. Because operators have limited information-processing capacity, an obstacle-warning system is needed to reduce their cognitive load during operation [40]. This function enables stable drone flight and effective obstacle avoidance. This study proposes an obstacle segmentation technique using U-disparity, V-disparity, and the scale obtained during 3D spatial reconstruction to detect only obstacles within a user-defined distance [41].

3.3.1 Obstacle Detection

U-disparity and V-disparity histograms can be generated using the MDE-based depth-information image estimated during 3D spatial reconstruction. As shown in Figure 3.6, U-disparity is a histogram representing the distribution of pixel values in the horizontal direction of the depth image. In regions where a particular depth value remains constant, a horizontal solid-line pattern appears in the histogram. Since obstacles tend to maintain a constant depth value across a horizontal region, the U-disparity histogram is useful for effectively identifying their horizontal extent and position. V-disparity represents the distribution of pixel values in the vertical direction of the depth image. Planar structures such as the floor generally exhibit a linear change in depth values and therefore appear as sloped lines in the histogram. In contrast, nonplanar structures such as obstacles appear as irregular patterns in the V-disparity histogram because their depth values change discontinuously or abruptly.

The obstacle segmentation process proceeds as follows. First, U-disparity and V-disparity histograms are generated and their principal patterns are detected. The discontinuities and slopes of the detected patterns are then analyzed to separate obstacles from the floor based on changes in depth values. Finally, the separated data are mapped back onto the display to visualize the position and size of obstacles, and the segmentation results are updated in real time to construct the display.

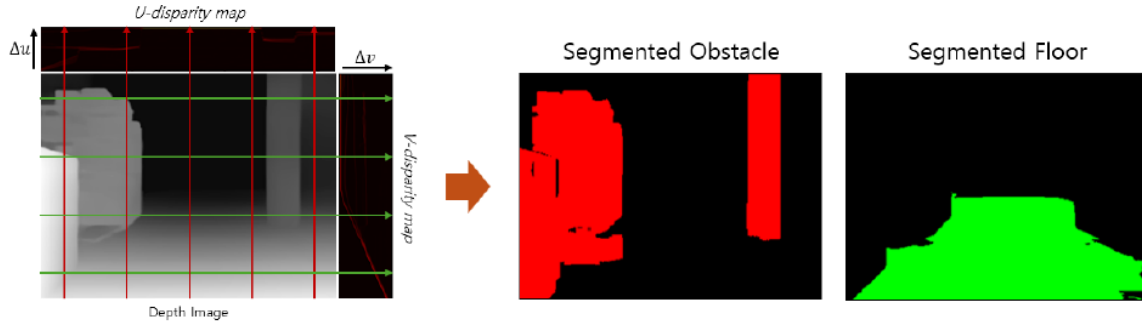


Figure 3.6. Obstacle and Floor Detection Using UV-Disparity

3.3.2 Morphological Operations and Obstacle Filtering

Noise and environmental factors during drone flight can cause pixels that do not correspond to actual obstacles to be detected. Such noise affects segmentation results and can ultimately degrade obstacle detection and avoidance performance. To address this, additional morphological operations can remove unnecessary regions and improve obstacle detection performance. As shown in Figure 3.7, the detected image was corrected using the closing technique, which consists of erosion and dilation operations [42]. Erosion removes small regions to eliminate unnecessary pixels, whereas dilation fills small holes or defects to reinforce continuous obstacle regions. This can remove small noise in the depth image and unnecessary regions that may occur along obstacle-detection boundaries.

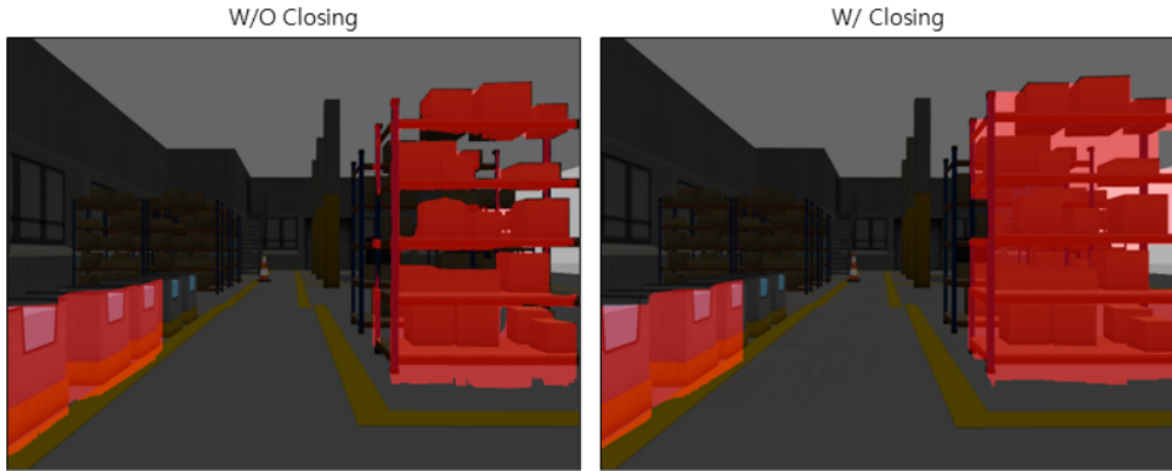


Figure 3.7. Segmentation Using Morphological Operations

Distance-based obstacle filtering is the process of using sensor data collected by the drone to distinguish nearby objects. This process is necessary to reduce cognitive load by selectively indicating the obstacles that an operator needs to consider in a disaster environment with a high obstacle density. The depth image generated by MDE for 3D spatial reconstruction provides an estimated depth value for each pixel, but these values do not include metric distance information and therefore require additional correction before direct use. To do so, the metric-distance-based scale calculated in Section 3.2.3 is applied to recalculate pixel values as metric distances. The recalculated distance values represent the actual distance between the drone and an obstacle, and only obstacles within a distance specified by the operator are filtered and displayed. As a result, only nearby obstacles can be shown on the display based on metric distances, as illustrated in Figure 3.8. This process allows the operator to adjust the obstacle-warning range flexibly and can therefore be used to adapt the display to the drone's flight environment.

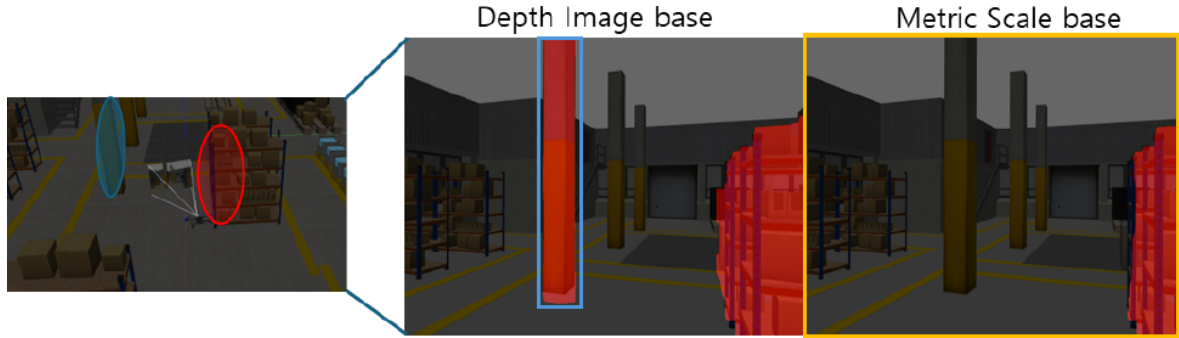


Figure 3.8. Image Segmentation Using a Relative-Distance-Based Depth Image and Nearby-Obstacle Segmentation Based on Metric Distance Data

3.4 Predicted Flight-Path Visualization

In teleoperation, an operator's trust in the system directly affects mission performance [43]. For drone operation, predicting the flight path and displaying it visually plays an important role in increasing trust in the system. This function analyzes the drone's current state and controller input data in real time to predict future motion, allowing the operator to understand the drone's direction of movement and path intuitively.

To implement this function, controller-command data are collected in real time and combined with IMU-sensor-based drone position and attitude data to calculate the predicted flight path. The calculated path is updated in real time by reflecting successive position-control commands relative to the drone's current position and is visualized as a curve. In particular, the concept of a vanishing point was applied to represent perspective when visualizing the flight path on a 2D display. The path gradually narrows along the drone's direction of movement, and more distant portions appear narrower and fainter. This perspective representation helps the operator intuitively understand the path's direction and sense of distance. As shown in Figure 3.9, paths can be displayed according to distance, along with left- and right-turning flight. The flight path is displayed in blue by default. This allows the operator to assess path safety visually with ease [44].

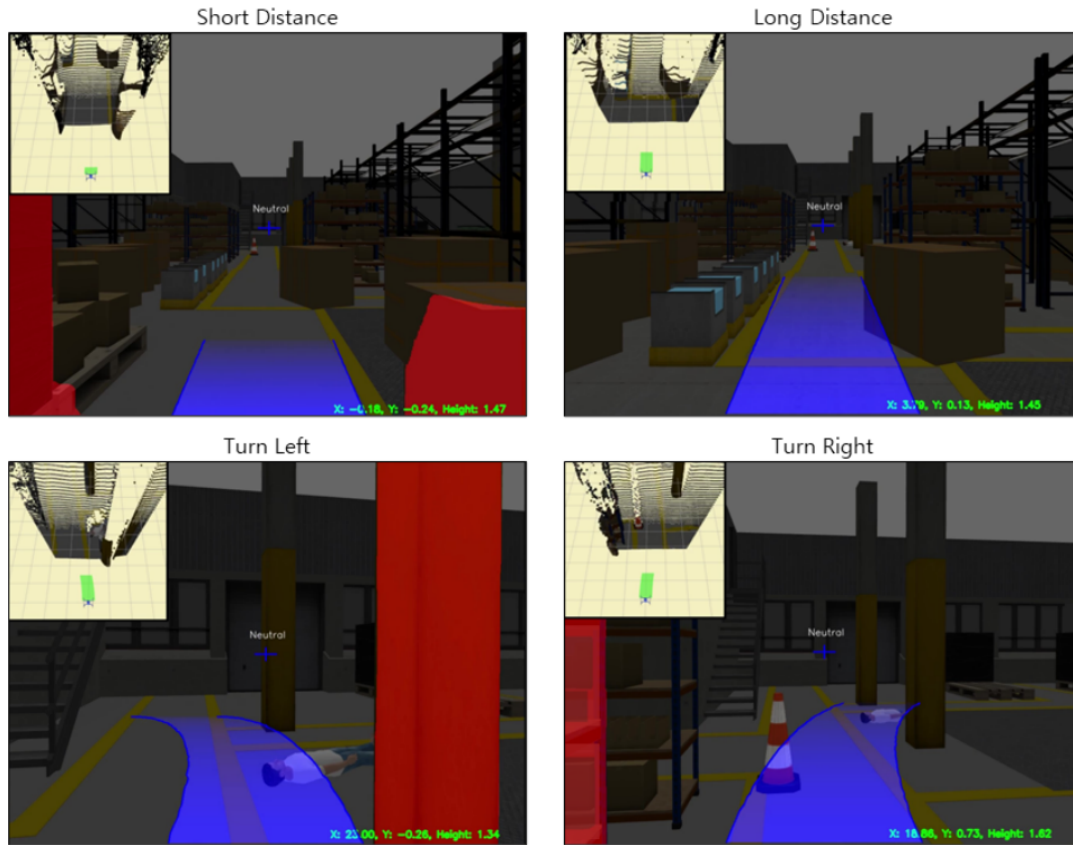


Figure 3.9. Predicted Flight-Path Interface

3.5 Integrated User Interface

The previously developed functions were integrated to form a new interface, as shown in Figure 3.10. This interface was configured to integrate the three main functions of 3D reconstructed spatial information, obstacle segmentation, and predicted-path visualization to enable drone operators to perform missions safely and effectively. Each function provides the operator with comprehensive and intuitive situational awareness.

The main display shows real-time video transmitted by the drone camera, with segmented obstacle images and predicted-path visualization overlaid on it. During flight, this allows the operator to see the drone's viewpoint directly while intuitively understanding nearby-obstacle information and the predicted flight path in real time. At the upper left of the display, a spatial reconstruction image visualizing the space ahead of the drone in three dimensions is arranged as a separate window. This helps the operator identify the drone's position within the flight space, recognize obstacle positions, and determine where flight is possible. The lower right displays the drone's current position, showing its relative position with respect to the flight's starting point in real time.

As a result, the 3D reconstructed spatial information visualizes the drone's surroundings in three dimensions, helping the operator clearly understand the flight situation and spatial context. Obstacle segmentation detects nearby obstacles in real time and highlights them visually, conveying the potential for collision in advance. Predicted-path visualization also calculates future motion from input data and presents the anticipated flight path intuitively through vanishing-point-based perspective representation and a color scheme. By combining these three functions, this integrated interface supports rapid operator responses to various situations, improves the safety and efficiency of drone operation to increase mission success, and reduces the operator's cognitive load.

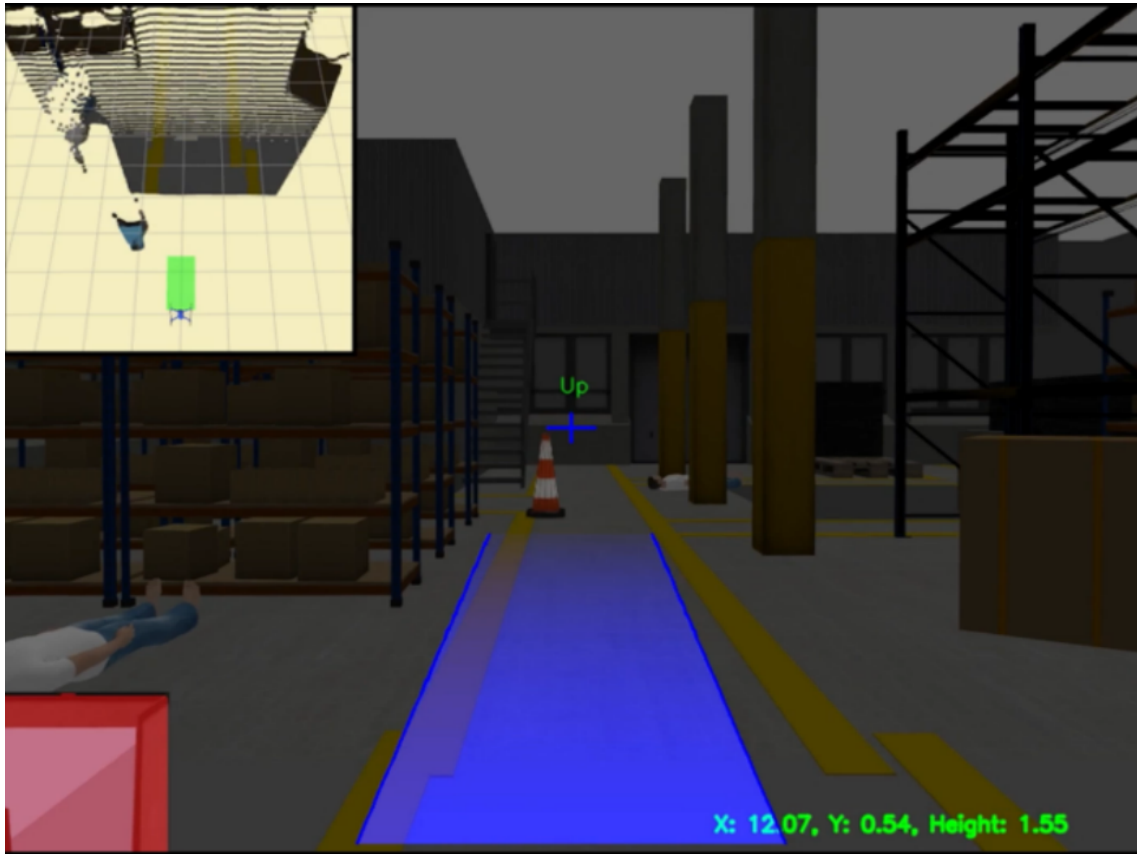


Figure 3.10. Integrated User Interface for Situational Awareness Assistance

Chapter 4. Experiments and Results

Chapter 4 describes three types of experiments conducted to verify the effectiveness of the controller and interface proposed in this study. In the first experiment, to validate the simulation experiments, a drone capable of indoor position estimation was constructed. The similarity between the simulation used in the experiments and the actual flight environment was compared, and operation of the developed system was checked. The second experiment was conducted to evaluate improvements in flight stability resulting from use of the proposed controller. Piloting experiments were performed in a simulation environment, and quantitative evaluation was conducted through trajectory analysis. Finally, to verify the situational-awareness assistance and cognitive-load level of the proposed user interface, the proposed and conventional interfaces were compared in a simulation environment and evaluated qualitatively using questionnaires.

4.1 Assessment of the Suitability of Simulation Experiments

The system-validation experiments were conducted with adults in their twenties who had no experience operating a drone. To ensure participant safety, a simulation-based experimental environment was configured to evaluate the performance of the proposed controller and user interface. To validate the simulation experiments, the flight performance of the physical drone was evaluated and its similarity to the simulation-based drone system was compared.

4.1.1 Fabrication of the Physical Drone

A drone capable of sharing the control system was built to reproduce the movement of the indoor-positioning drone in the same way as in the simulation. The main objectives of fabricating the drone were to implement the control system verified during the simulation stage in a real environment and to compare and analyze the drone's operating characteristics against simulation data. The drone is controlled using the Robot Operating System (ROS) to enable remote data transmission, reception, and control, and was designed as a quadcopter with four motors. The specifications and appearance of the fabricated drone are shown in Table 4.1 and Figure 4.1, respectively.

Table 4.1. Specifications of the fabricated drone

Feature	Description
Type	Quadrotor
Frame Diameter	380 mm
Height	185 mm
Propeller Diameter	213 mm
Motor Thrust	980KV
Flight Mode	Manual Control Autonomous Positioning

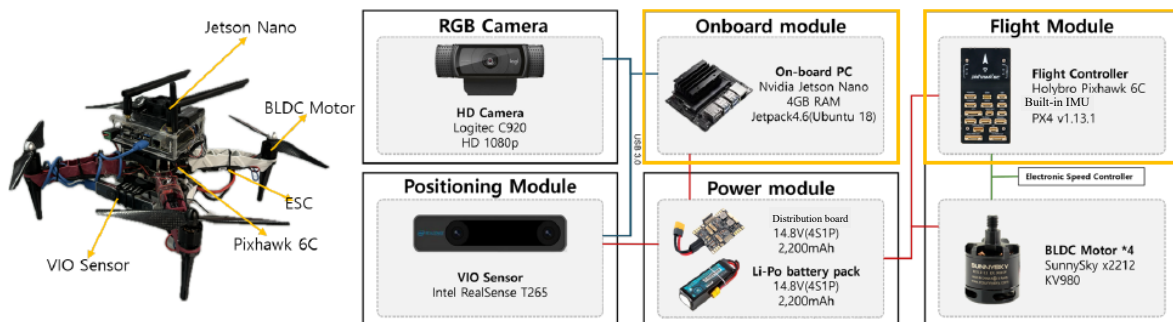


Figure 4.1. Fabricated drone

An NVIDIA Jetson Nano was used as the drone's onboard PC, running the Ubuntu 18.04-based JetPack 4.6 operating system. The Jetson Nano is compact yet provides a high-performance GPU suitable for image processing and real-time data communication. It was configured in the ROS Melodic environment to transmit and receive control commands, drone data, and image data remotely. A Holybro Pixhawk 6C was used as the drone's flight controller. The Pixhawk 6C is an open-source platform operating on the PX4-Autopilot system. It uses an STM32H7 processor for high-speed computation and includes two IMU sensors to support attitude stability and position control during flight. The specifications of the Pixhawk 6C are provided in Table 4.2. This controller is connected to the onboard PC via UART communication to exchange the drone's position information and target-position data.

Table 4.2. Pixhawk 6C specifications

Holybro Pixhawk 6C		
Photograph	Item	Specification
	FMU Processor	STM32H743
	Sensor Configuration	ICM-42688-P, BMI055
	Dimensions	84.8 mm X 44 mm X 12.4 mm
	Weight	59.3 g

A tracking camera was mounted on the underside of the drone to enable position control in indoor environments using Visual Inertial Odometry (VIO). The installed camera is an Intel T265, connected to the Jetson Nano through a USB 3.0 interface to provide accurate position-estimation data. The specifications of this tracking camera are provided in Table 4.3.

A Logitech C922 webcam was used to transmit forward-facing images. This camera can capture images with a 78-degree field of view and is connected to the Jetson Nano mounted on the drone through a USB 2.0 cable. The specifications of the C922 camera are provided in Table 4.4.

Table 4.3. T265 tracking camera specifications

Intel RealSense Tracking Camera T265		
Photograph	Item	Specification
	Dimensions	108 mm x 25 mm x 13 mm
	Weight	55 g
	Sensor Configuration	1 IMU, 2 Fisheye lenses
	Diagonal Field of View (FOV)	163±5°

Table 4.4. C922 webcam specifications

C922 PRO HD STREAM WEBCAM		
Photograph	Item	Specification
	Dimensions	95 mm X 71 mm X 44 mm
	Resolution	720p/60fps
	Diagonal Field of View (dFOV)	78°

A 14.8 V (4S), 2200 mAh, 45C lithium-polymer battery was installed to supply power to the drone. A distribution board was placed at the center of the airframe to provide stable power and support the stable operation of the drone's motors, flight controller, and onboard PC.

The drone fabrication process consisted of frame assembly, hardware integration, and software configuration.

The drone frame consists of a lightweight, durable fiberglass central plate and plastic motor arms, and was designed to accommodate the onboard PC and battery at its center. Each component was mounted on the frame, and UART communication was configured through GPIO connections for communication between the onboard PC and the flight controller. The tracking camera and forward-facing camera were each connected to the onboard PC through USB cables. The drone control system and VIO-based position-control algorithm were then integrated in the ROS Melodic environment, and the system was configured to transmit and receive data remotely. Hovering and manual-control tests were conducted on the fabricated drone to check the operation of each component, and stability was verified through repeated indoor flight experiments.

4.1.2 Evaluation of the Physical Drone's Motion

Experiments with a physical drone sharing the same control system as the simulation were conducted to verify the suitability of simulation experiments for evaluating the controller and user interface. First, to verify the accuracy of the VIO system using the tracking camera mounted on the drone, the airframe was moved from the starting position (0, 0) to (1, 0), (2, 0), (2, 1), and (2, -1), as shown in Figure 4.2. The positions estimated using VIO were measured, and the errors were calculated. 8 repeated experiments showed distance errors of 5.25% in the x direction and 3.21% in the y direction relative to the starting position.

Next, the drone-control and remote image-transmission and reception capabilities using the proposed controller were verified. When the controller system to be used in the experiments was applied identically to the physical drone, the drone was confirmed to be controlled in the same way as in the simulation. In addition, the images captured by the drone and the controller's control data were used to check whether the functions of the proposed interface had been implemented. All functions were confirmed to operate normally, as shown in Figure 4.3.

These experiments confirmed that indoor remote control and image transmission and reception operated smoothly on a physical drone using the same control system as the simulation environment, and verified that the simulation experiments reflect the real environment.

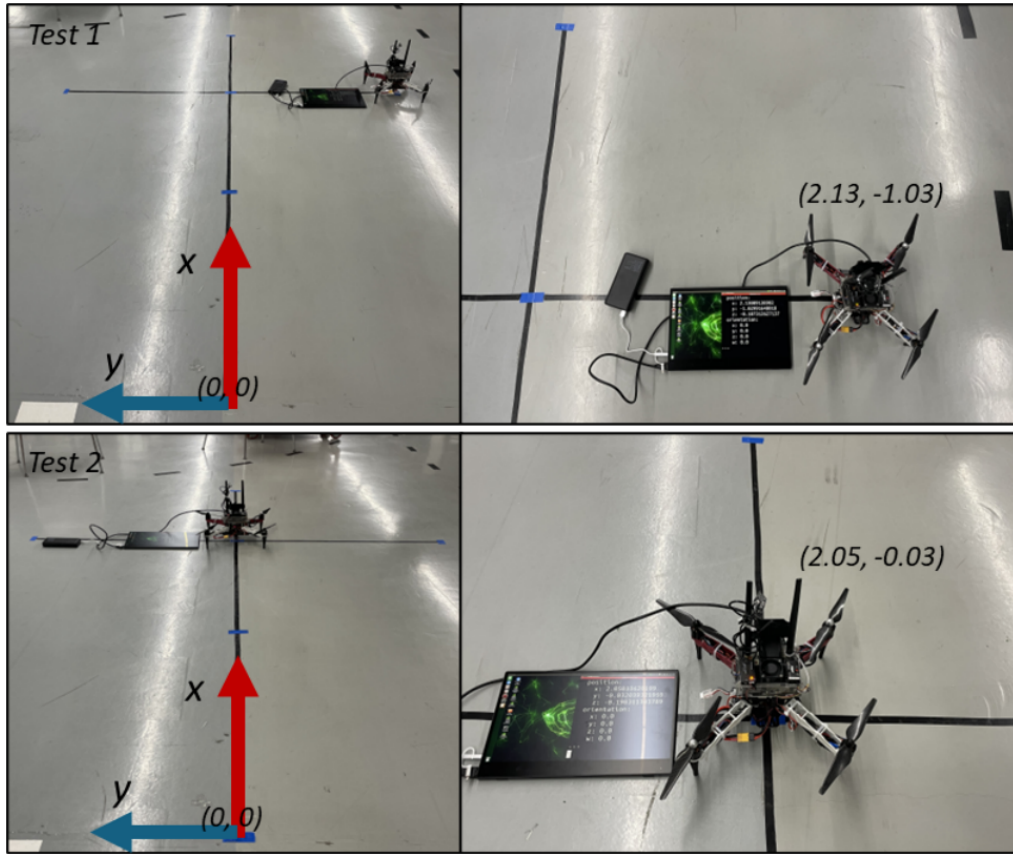


Figure 4.2. Verification of position-estimation performance of the physical drone

4.2 Controller Performance Evaluation

The controller-performance evaluation experiment was conducted to evaluate flight performance by comparing the controller proposed in this study with the gamepad-type controller used in existing drone systems. The experiment was conducted in a Gazebo-based simulation environment, and performance was evaluated according to quantitative criteria.

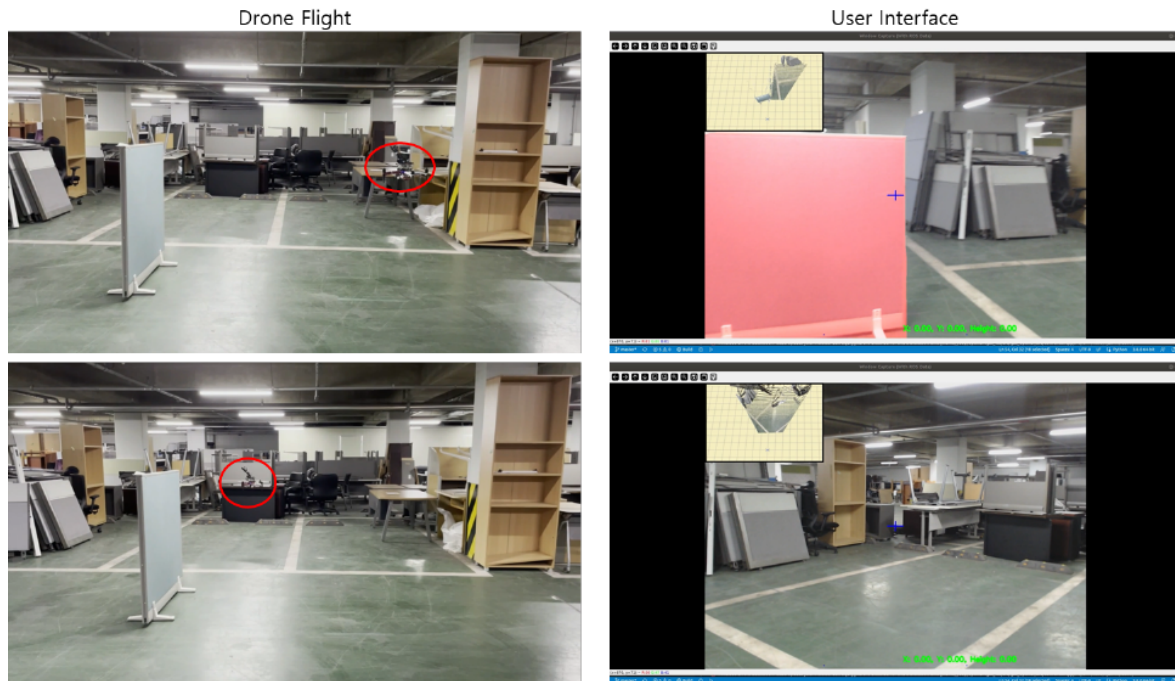


Figure 4.3. Verification of remote control and image transmission and reception of the physical drone

4.2.1 Controller Performance Evaluation Scenario

To evaluate controller performance, a flight scenario in which the drone passes through square gates was established in a Gazebo-based simulation environment, as shown in Figure 4.4. The scenario consisted of a mission to pass sequentially through 7 square gates arranged counterclockwise on a plane measuring 25 m by 25 m. Each square gate measured 1.4 m in width and height, and the total altitude variation of the arranged gates was 3.0 m. The experiment used the existing RGB-based user interface, and all variables other than the controller were kept identical.

The experiment involved 6 participants: 4 men and 2 women in their twenties with no experience operating drones. Before the experiment, each participant performed a one-minute trial flight to become familiar with the controller. Each then completed the gate-passing flight once with each controller. The drone's attitude and trajectory during flight were recorded for analysis of the results.

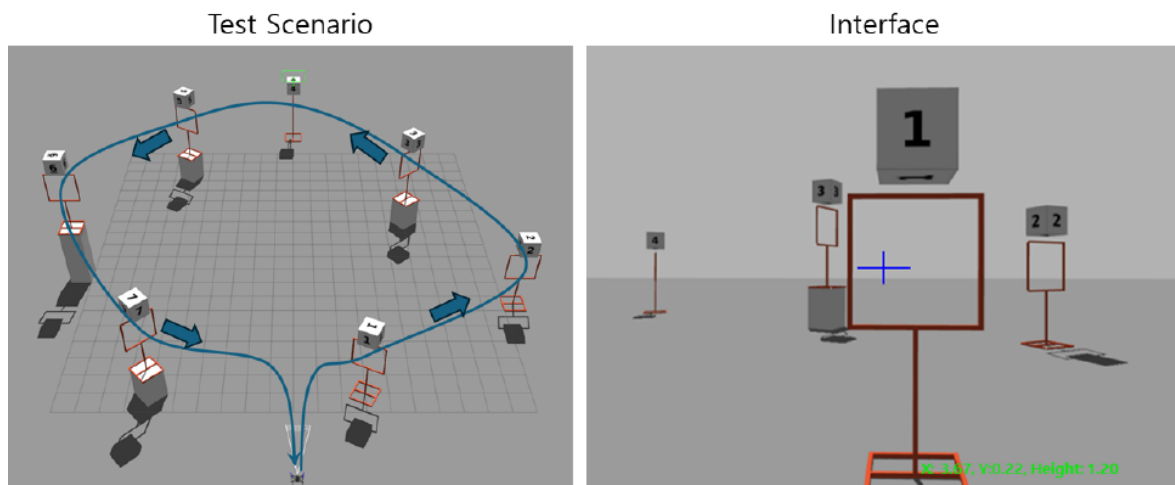


Figure 4.4. Controller experiment scenario and user interface

4.2.2 Controller Experiment Results

The flight trajectories of all participants are presented in Figure 4.5, plotted in three dimensions and separated into the gamepad and proposed-controller conditions. Flight performance was evaluated using two metrics, Path Smoothness and

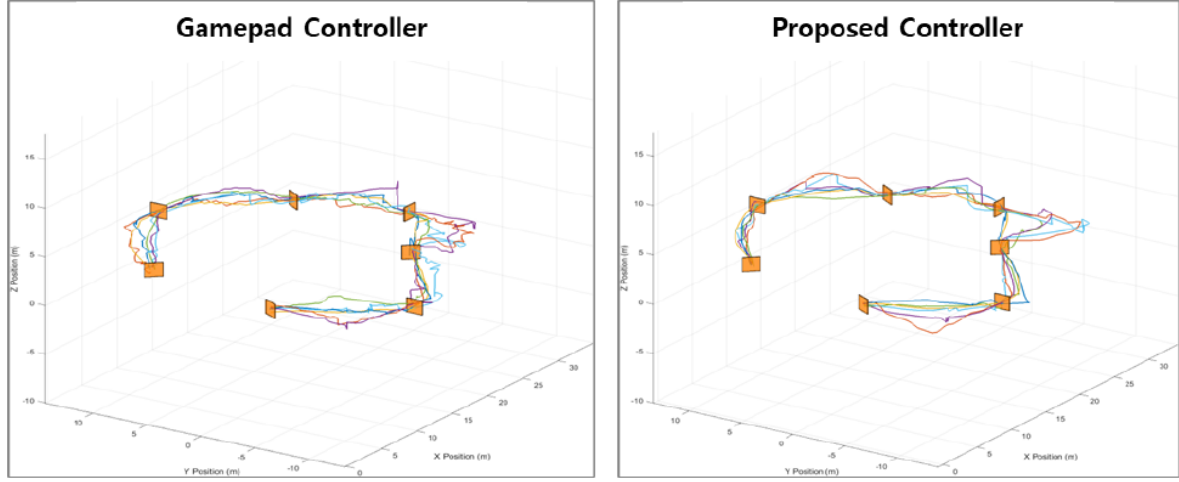


Figure 4.5. Flight trajectories in the controller experiment

Minimum Distance from Gates. The respective equations are as follows.

$$d = \sqrt{(x_t - x_j)^2 + (y_t - y_j)^2 + (z_t - z_j)^2}$$

$$d = \sqrt{(x_t - x_j)^2 + (y_t - y_j)^2 + (z_t - z_j)^2}$$

Path Smoothness is a metric used to evaluate the degree of stable flight by calculating unnecessary movements that occur during travel. A lower path-smoothness value is considered to indicate more effective travel [45]. Minimum Distance from Gates is a metric introduced to verify stability when passing through obstacles by measuring how close the airframe's trajectory is to the center of each gate. This metric was used because flying through safe spaces is important when navigating indoor disaster environments that require passage through narrow spaces. The distance between the center of a gate and the airframe's trajectory was measured to verify the controller's effectiveness in passing through obstacles.

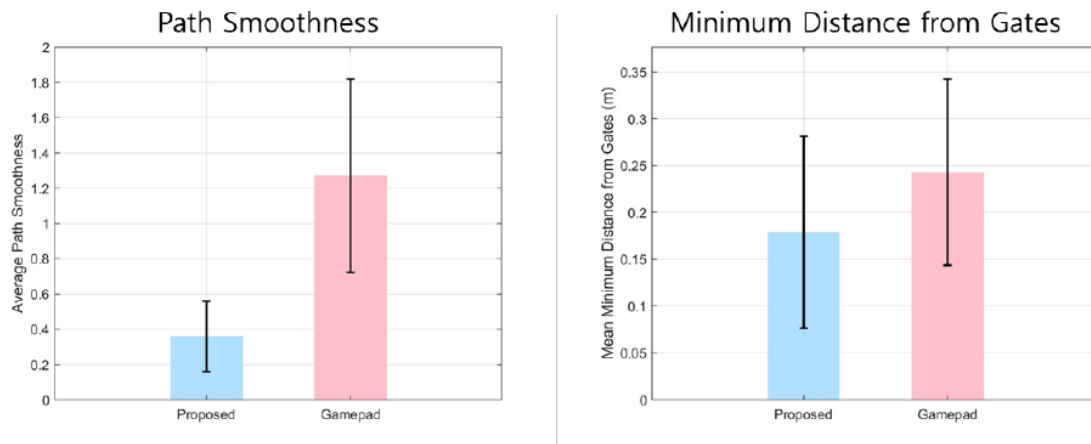


Figure 4.6. Results of the controller comparison experiment

As shown in Figure 4.6, the experimental results indicated that, when the proposed controller was used, the Path Smoothness value was 0.358, confirming improved path smoothness compared with the gamepad controller's value of 1.2712. This shows that the proposed controller improves the airframe's travel stability and enables efficient flight with fewer unnecessary movements. Improved performance was also observed for Minimum Distance from Gates. The proposed controller yielded a value of 0.18, compared with 0.24 for the gamepad controller, indicating that it maintained a trajectory closer to the center of the gates. This means that the proposed controller reduces the possibility of collisions in narrow spaces and supports stable flight.

Table 4.5. Comparison of experimental results for the proposed controller and the gamepad-type controller

	Proposed	Gamepad
Mean	0.358	1.2712
SD	0.20001	0.54762

In conclusion, the controller proposed in this study demonstrated improved flight-trajectory stability and smoothness compared with the gamepad. These results mean that the proposed controller is more effective for performing flight missions in narrow and complex spaces such as indoor disaster environments.

4.3 User Interface Performance Evaluation

The user-interface performance evaluation experiment was conducted to compare the user interface proposed in this study with the RGB-image-based interface used in existing drone systems. The experiment was performed in a Gazebo simulation environment and aimed to verify whether the interface improved the usability and workload of the drone teleoperation system.

4.3.1 User Interface Performance Evaluation Scenario

To evaluate user-interface performance, the image-based user interface of an existing drone system was compared with the user interface proposed in this study, as shown in Figure 4.7. The experiment was performed in the Gazebo-based simulation environment configured as shown in Figure 4.8. Participants were required to perform a mission to search for 4 people to be rescued, positioned randomly within a factory.

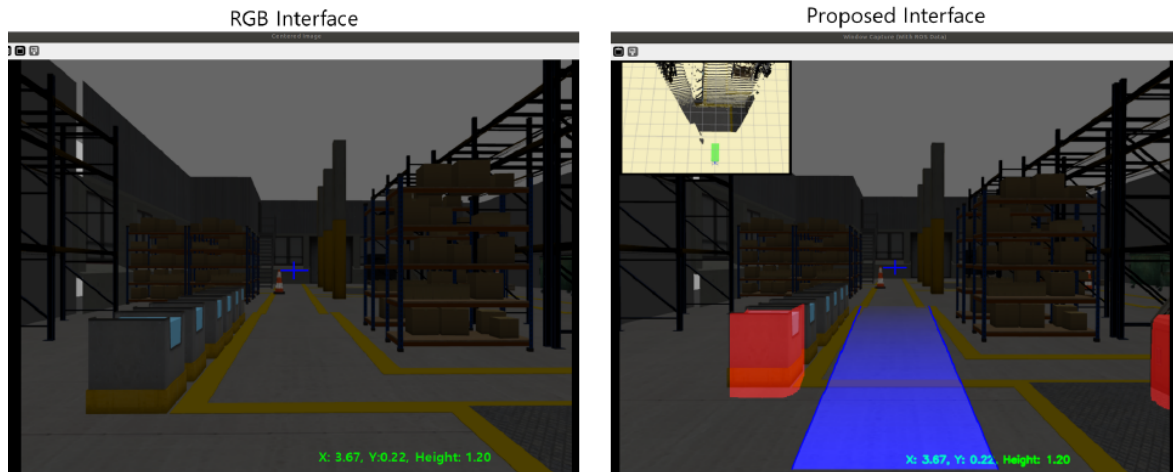


Figure 4.7. Comparison conditions for the user-interface experiment

The experiment involved 6 participants: 4 men and 2 women in their twenties with no experience operating drones. All participants used the same controller and conducted one flight experiment with the proposed user interface and one with the existing interface. After the experiments, participants completed two questionnaires to evaluate the interfaces' usability and workload.

4.3.2 User Interface Experiment Results

Interface performance was compared through user evaluations, using the results of the System Usability Scale (SUS) questionnaire shown in Figure 4.9 and the NASA-TLX questionnaire shown in Figure 4.10, administered after the flight experiments

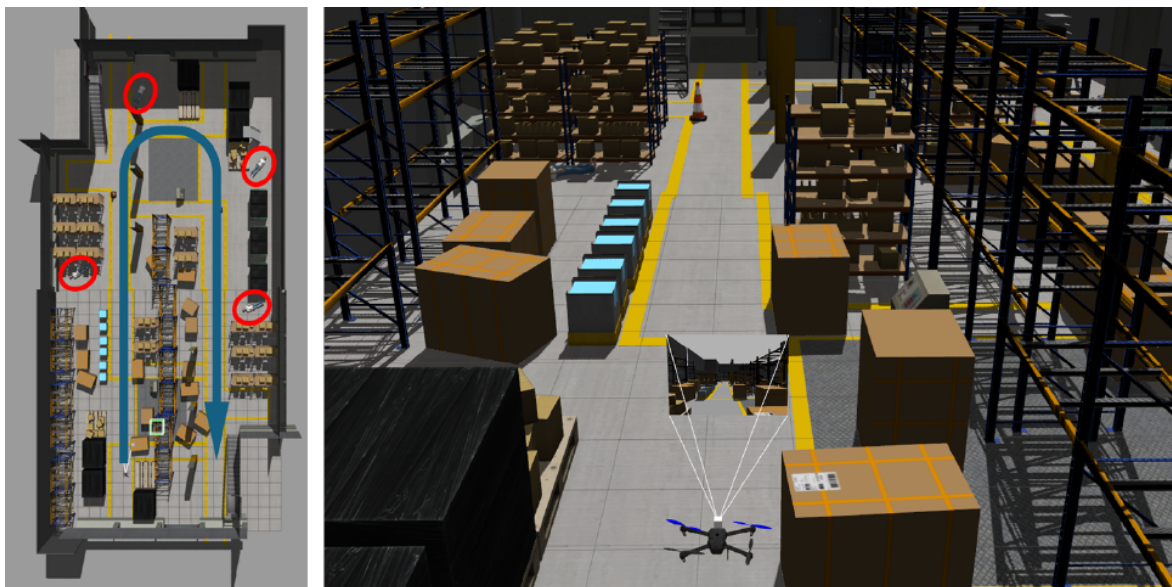


Figure 4.8. User-interface experiment scenario

[46, 47]. The first questionnaire, the System Usability Scale (SUS), is used to evaluate the usability of a developed system and consists of 10 items. Each item is rated on a 5-point scale ranging from "strongly agree" to "strongly disagree." These ratings are converted into an overall score out of 100, with a higher score indicating better system usability. The second questionnaire, the NASA-TLX (Task Load Index), is used to measure the cognitive load experienced by a user while using a system. This questionnaire evaluates 6 workload components experienced during task performance: mental demand, physical demand, temporal

demand, performance, effort, and frustration. Each component is measured on a scale from 0 to 100 based on the user's subjective evaluation. A lower score indicates lower cognitive load when using the system. This allows the effects of the task environment on the user's cognitive and physical resources to be analyzed.

As shown in Figure 4.11, the experimental results indicated that the proposed user interface performed better than the existing RGB-image-based user interface. The SUS score for the proposed interface was 80, compared with 62.5 for the existing interface, demonstrating improved usability of the proposed interface. In the NASA-TLX results, the proposed interface also achieved better scores in all categories. This means that, when using the proposed interface, pilots feel more comfortable with the system and experience less cognitive load during mission execution than when using the existing interface.

	Strongly disagree					Strongly agree
1. I think that I would like to use this ballot frequently.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
2. I found the ballot unnecessarily complex.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
3. I thought the ballot was easy to use.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
4. I think that I would need the support of a poll official to be able to use this system.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
5. I found the various parts of this ballot were well integrated.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
6. I thought there was too much inconsistency in this ballot.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
7. I would imagine that most people would learn to use this ballot very quickly.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
8. I found the ballot very awkward to use.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
9. I felt very confident using the ballot.	☐	☐	☐	☐	☐	
	1	2	3	4	5	
10. I needed to learn a lot of things before I could get going with this ballot.	☐	☐	☐	☐	☐	
	1	2	3	4	5	

Figure 4.9. System Usability Scale (SUS) questionnaire

Name	Task	Date
------	------	------

Mental Demand How mentally demanding was the task?

Very Low Very High

Physical Demand How physically demanding was the task?

Very Low Very High

Temporal Demand How hurried or rushed was the pace of the task?

Very Low Very High

Performance How successful were you in accomplishing what you were asked to do?

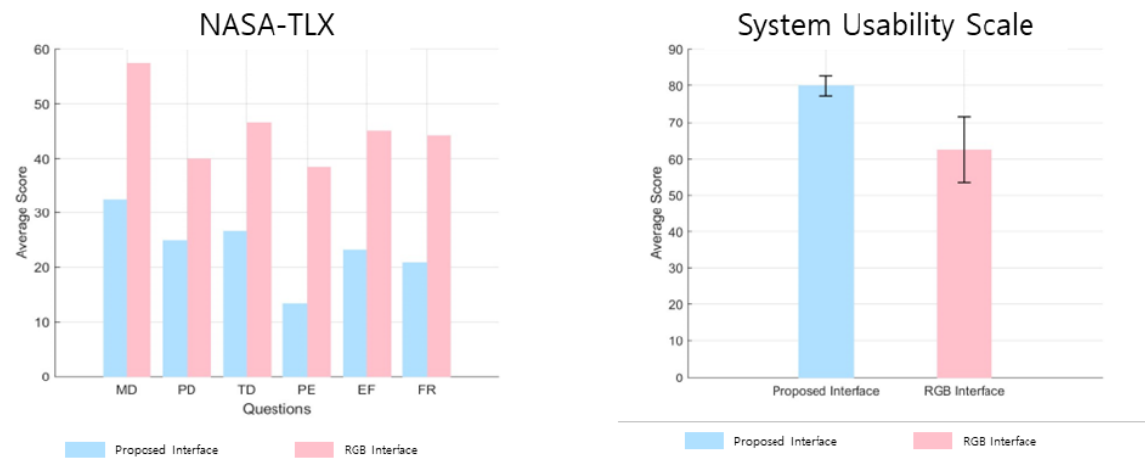
Perfect Failure

Effort How hard did you have to work to accomplish your level of performance?

Very Low Very High

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low Very High

Figure 4.10. NASA-TLX questionnaire**Figure 4.11.** User-interface experiment results**Table 4.6.** Results of the SUS and NASA-TLX questionnaires

	SUS	NASA-TLX					
		MD	PD	TD	PE	EF	FR
Proposed Interface	80	32.5	25	26.7	13.3	23.3	20.8
RGB Interface	62.5	57.5	40	46.7	38.3	45	44.17

In conclusion, the user interface proposed in this study was confirmed to improve ease of use and task efficiency compared with the interface of the existing system. This demonstrates that the use of an integrated user interface is effective for drone operation in indoor disaster environments.

Chapter 5. Conclusion

In this study, a controller and a monocular-camera-based user interface were designed for drones responding to indoor disasters, and were verified in a simulation environment similar to actual drone flight. The proposed controller and user interface were designed to respond effectively to the constraints of indoor environments that can arise during drone teleoperation. An intuitive controller and an integrated user interface supporting situational awareness were thereby developed to enable stable and efficient flight even inside complex structures. The experimental results confirmed that the proposed controller substantially improved path smoothness compared with the gamepad-type controller used in existing drone systems, and increased the accuracy of passing through the center of a gate by 25%. In addition, experiments in which participants performed a mission to search for people to be rescued using the user interface showed that the overall SUS score of the proposed user interface improved by 28% compared with the existing RGB-image-based user interface. In the NASA-TLX questionnaire used to measure workload, improvements of 43%, 38%, 43%, 65%, 48%, and 53% were confirmed for mental demand (MD), physical demand (PD), temporal demand (TD), performance (PE), effort (FE), and frustration (FR), respectively.

The proposed controller improved control stability through intuitive control mapping, while the new user interface contributed to greater usability and reduced workload by effectively providing pilots with the information needed for remote flight. The experimental evaluation demonstrated that these improvements substantially enhance drone travel efficiency and stability, and can meaningfully improve pilots' mission-performance capabilities in spatial exploration and searches for people to be rescued in indoor disaster environments.

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